



Red Hat OpenShift AI Self-Managed 3.0

Working with the model catalog

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Abstract

Discover and evaluate generative AI models in the model catalog and select models to register, deploy, and customize.

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PREFACE

As a data scientist or AI engineer in OpenShift AI, you can discover and evaluate the generative AI models that are available in the model catalog. From the model catalog, you can select the models that you want to register, deploy, and customize.

CHAPTER 1. OVERVIEW OF THE MODEL CATALOG AND MODEL REGISTRIES

The model catalog provides a curated library where data scientists and AI engineers can discover and evaluate the available generative AI (gen AI) models to find the best fit for their use cases.

A model registry acts as a central repository for administrators, data scientists, and AI engineers to register, version, and manage the lifecycle of AI models before configuring them for deployment. A model registry is a key component for AI model governance.

1.1. MODEL CATALOG

Data scientists and AI engineers can use the model catalog to discover and evaluate the gen AI models that are available and ready for their organization to register, deploy, and customize.

The model catalog provides models from different providers that you can search, discover, and evaluate before you register models in a model registry and deploy them to a model serving runtime. Third-party gen AI models are benchmarked by Red Hat for performance and quality by using open-source evaluation datasets. You can compare performance metrics for specific hardware configurations and determine the most suitable option for deployment.

OpenShift AI provides a default model catalog, which includes models from providers such as Red Hat, IBM, Meta, Nvidia, Mistral AI, and Google. OpenShift AI administrators can configure the available repository sources for models displayed in the model catalog.

For more information about how data scientists and AI engineers can use the model catalog, see [Working with the model catalog](#).

1.2. MODEL REGISTRY

A model registry is an important component in the lifecycle of an artificial intelligence/machine learning (AI/ML) model, and is a vital part of any machine learning operations (MLOps) platform or workflow. A model registry acts as a central repository, storing metadata related to machine learning models from development to deployment. This metadata ranges from high-level information like the deployment environment and project, to specific details like training hyperparameters, performance metrics, and deployment events.

A model registry acts as a bridge between model experimentation and serving, offering a secure, collaborative metadata store interface for stakeholders in the ML lifecycle. Model registries provide a structured and organized way to store, share, version, deploy, and track models.

OpenShift AI administrators can create model registries in OpenShift AI and grant model registry access to data scientists and AI engineers. For more information, see [Managing model registries](#).

Data scientists and AI engineers with access to a model registry can use it to store, share, version, deploy, and track models. For more information, see [Working with model registries](#).

CHAPTER 2. DISCOVERING AND EVALUATING MODELS IN THE MODEL CATALOG

You can discover and evaluate the available gen AI models in the model catalog to find the best fit for your use cases. You can select from available model categories, search by text, and filter by labels.

For validated models, you can view performance benchmark data for specific hardware configurations to evaluate and compare options for deployment.

Prerequisites

- You are logged in to Red Hat OpenShift AI.
- The model registry component is enabled in your OpenShift AI deployment. For more information, see [Enabling the model registry component](#).

Procedure

1. From the OpenShift AI dashboard, click **AI hub** → **Catalog**.
2. The **Model Catalog** page provides a high-level view of available models, including the model category, name, description, and labels such as task, license, and provider. You can also view performance benchmarks for validated models from third parties.
3. In the menu bar, select from the available model categories:
 - **All models**: All models available in the model catalog.
 - **Red Hat AI models**: Models provided and supported by Red Hat.
 - **Red Hat AI validated models**: Third-party models benchmarked by Red Hat for performance and quality by using open-source evaluation datasets.
 - **Community and custom models**: Additional third-party and community models configured by your administrator.
4. You can use the search bar to find a model in the catalog. Enter text to search by model name, description, or provider.
5. You can use the filter menu to search and select filters by the following labels:
 - **Task**: For example, **Text-generation**.
 - **Provider**: For example, **Meta**.
 - **License**: For example, **Apache 2.0**.
6. Click the name of a model to view the model details page. This page displays the model description and the **Model card** information supplied by the model provider. This includes details such as the model's intended use and potential limitations, training parameters and datasets, and evaluation results.
7. For validated models, click the **Performance Insights** tab to view performance benchmark data to compare performance metrics for specific hardware configurations and to determine the most suitable options for deployment.
You can filter the performance data by the following options:

- **Workload type:** Select a workload type to view performance under different token lengths, for example, **Chatbot**.
 - **Max latency:** Set your maximum acceptable latency. Hardware configurations that respond slower than this value are hidden. You can select a specific metric in the list:
 - **E2E** (end-to-end request latency): The time taken from submitting the request to receiving the final response.
 - **TTFT** (time to first token): The time that the user must wait before seeing output from the model.
 - **TPS** (tokens per second): The total number of tokens that are output per second.
 - **ITL** (inter-token latency): The average time taken between consecutive tokens. You can also select a percentile value, for example, **P90**.

Use the slider to set the maximum acceptable latency value in milliseconds, and click **Apply filter**.
 - **Min RPS:** Set to only show models that can handle at least this number of requests. Hardware configurations that perform below this value are hidden. Use the slider to set the minimum requests per second value, and click **Apply filter**.
 - **Hardware type:** Select one or more hardware types from the list, for example, **H200**. You can click **Clear all filters** to reset your filters and try again.
8. For categories with more than 10 models, you can click **Load more models** to scroll and view additional models available in the catalog. Repeat this step until all models are loaded.

Verification

- For all models, you can view the information about a selected model on the model details page.
- For validated models, you can view the benchmark information about a selected model on **Performance Insights** tab.

CHAPTER 3. REGISTERING A MODEL FROM THE MODEL CATALOG

As a data scientist or AI engineer, you can register models directly from the model catalog and create the first version of the new model.

Prerequisites

- You are logged in to Red Hat OpenShift AI.
- You have access to an available model registry in your deployment.

Procedure

1. From the OpenShift AI dashboard, click **AI hub** → **Catalog**.
2. The **Model Catalog** page provides a high-level view of available models, including the model category, name, description, and labels such as task, license, and provider.
3. You can use the search bar to search by model name, description, or provider.
4. You can use the filter menu to search and select filters by task, provider, or license.
5. Click the name of a model to view the model details page.
6. Click **Register model**.
7. From the **Model registry** drop-down list, select the model registry that you want to register the model in.
8. In the **Model details** section, configure details to apply to all versions of the model:
 - a. Optional: In the **Model name** field, update the name of the model.
 - b. Optional: In the **Model description** field, update the description of the model.
9. In the **Version details** section, enter details to apply to the first version of the model:
 - a. In the **Version name** field, enter a name for the model version.
 - b. Optional: In the **Version description** field, enter a description for the first version of the model.
 - c. In the **Source model format** field, enter the name of the model format, for example, **ONNX**.
 - d. In the **Source model format version** field, enter the version of the model format.
10. In the **Model location** section, the URI of the model is displayed.
11. Click **Register model**.

Verification

- The new model details and version are displayed on the **Overview** tab on the model details page.

- The new model and version are displayed on the **Model registry** page.

CHAPTER 4. DEPLOYING A MODEL FROM THE MODEL CATALOG

You can deploy models directly from the model catalog.



NOTE

OpenShift AI model serving deployments use the global cluster pull secret to pull models in ModelCar format from the catalog.

For more information about using pull secrets in OpenShift, see [Updating the global cluster pull secret](#) in the OpenShift documentation.

Prerequisites

- You have completed the prerequisites in [Deploying models](#).
- The model registry component is enabled in your OpenShift AI deployment. For more information, see [Enabling the model registry component](#).

Procedure

1. From the OpenShift AI dashboard, click **AI hub** → **Catalog**.
2. The **Model Catalog** page provides a high-level view of available models, including the model category, name, description, and labels such as task, license, and provider.
3. You can use the search bar to search by model name, description, or provider.
4. You can use the filter menu to search and select filters by task, provider, or license.
5. Click the name of a model to view the model details page.
6. Click **Deploy model** to display the **Deploy model** dialog.
7. From the **Project** drop-down list, select a project in which to deploy your model.
8. In the **Model deployment** section:
 - a. Optional: In the **Model deployment name** field, enter a unique name for your model deployment. This field is autofilled with a value that contains the model name by default. This is the name of the inference service created when the model is deployed.
 - b. Optional: Click **Edit resource name**, and then enter a specific resource name for the model deployment in the **Resource name** field. By default, the resource name matches the name of the model deployment.



IMPORTANT

Resource names are what your resources are labeled as in OpenShift. Your resource name cannot exceed 253 characters, must consist of lowercase alphanumeric characters or -, and must start and end with an alphanumeric character. Resource names are not editable after creation.

The resource name must not match the name of any other model deployment resource in your OpenShift cluster.

- c. From the **Serving runtime** list, select a model-serving runtime that is installed and enabled in your OpenShift AI deployment. If project-scoped runtimes exist, the **Serving runtime** list includes subheadings to distinguish between global runtimes and project-scoped runtimes.
- d. From the **Model framework** list, select a framework for your model.



NOTE

The **Model framework** list shows only the frameworks that are supported by the model-serving runtime that you specified when you deployed your model.

9. From the **Deployment mode** list, select **KServe RawDeployment**.
 - a. In the **Number of model server replicas to deploy** field, specify a value.
 - b. From the **Model server size** list, select a value.
 - c. If you have created a hardware profile, select a hardware profile from the **Hardware profile** list. If project-scoped hardware profiles exist, the **Hardware profile** list includes subheadings to distinguish between global hardware profiles and project-scoped hardware profiles.
 - d. In the **Model route** section, select the **Make deployed models available through an external route** checkbox to make your deployed models available to external clients.
 - e. In the **Token authentication** section, select the **Require token authentication** checkbox to require token authentication for your model server. To finish configuring token authentication, perform the following actions:
 - i. In the **Service account name** field, enter a service account name for which the token will be generated. The generated token is created and displayed in the **Token secret** field when the model server is configured.
 - ii. To add an additional service account, click **Add a service account** and enter another service account name.
10. In the **Source model location** section, select **Current URI** to deploy the selected model from the catalog.
11. Optional: Customize the runtime parameters in the **Configuration parameters** section:
 - a. Modify the values in **Additional serving runtime arguments** to define how the deployed model behaves.
 - b. Modify the values in **Additional environment variables** to define variables in the model's environment.

12. Click **Deploy**.

Verification

- The model deployment is displayed on the **AI hub** → **Deployments** page.
- The model deployment is displayed in the **Latest deployments** section of the model details page.
- The model deployment is displayed on the **Deployments** tab for the model version.

Additional resources

- [Deploying models on the model serving platform](#).

CHAPTER 5. CONFIGURING MODEL CATALOG SOURCES IN OPENSIFT

As an OpenShift cluster administrator, you can configure a custom model catalog source by adding an entry to the **model-catalog-sources** config map in OpenShift. This entry provides details such as the catalog name and the location of the catalog source file that specifies your model definitions.

Prerequisites

- You are logged in to the OpenShift web console with cluster administrator privileges.
- You have created a catalog source file that specifies your model definitions. For an example catalog source file, see the [sample-catalog.yaml](#) file from Kubeflow Model Registry.

Procedure

1. In the OpenShift web console navigation menu, click **Workloads → ConfigMaps**.
2. In the **Name** text box, enter **model-catalog-sources**.
3. Click the **model-catalog-sources** config map name.
4. Click the **YAML** tab to view the contents of the config map **.yaml** file.
5. Find the **data: sources.yaml** section to add custom catalog sources.
The following sample shows the default **sources.yaml** section in the **model-catalog-default-sources** config map as an example:

```
data:
  sources.yaml: |-
    catalogs:
      - name: Red Hat AI
        id: redhat_ai_models
        type: yaml
        properties:
          yamlCatalogPath: /shared-data/models-catalog.yaml
        labels:
          - Red Hat AI
      - name: Red Hat AI validated
        id: redhat_ai_validated_models
        type: yaml
        properties:
          yamlCatalogPath: /shared-data/validated-models-catalog.yaml
        labels:
          - Red Hat AI validated
```

6. In the **model-catalog-sources** config map, add a new entry under the **catalogs** field and provide the following details:
 - **name:** The user-friendly name of the catalog source, for example, **My Sample Catalog**.
 - **id:** The unique ID of the catalog source, for example: **my-sample-catalog**.
 - **type:** The catalog source type. The only supported type is **yaml**.

- **properties.yamlCatalogPath:** The location of your catalog source file, for example: **my-sample-catalog.yaml**.
- **labels:** Optional labels for your catalog source, for example: **My Sample Catalog**.

7. Click **Save**.

Verification

1. In the OpenShift AI dashboard, click **AI hub → Catalog**.
2. In the menu bar, to view models from your new catalog source:
 - If your catalog source has no label, click **Community and custom models**
 - If your catalog source has a label, click **<your-label-name> models**.



NOTE

It might take a few minutes for the changes to be reflected on the dashboard.

Additional resources

- [Kubeflow Model Registry documentation on configuring catalog sources](#)